REVWORKFORCE

Human Resource Management

Project Duration: 20-Feb-2026 to 11-Mar-2026

2 0 2 6

by: **Revature CTS TRNG-00002359**

Hello,

We're RevWorkForce Team

Revature CTS TRNG-00002359

1. Mastan Sayyad (Team Lead)
2. Gururaj Shetty (UI Lead/BK)
3. Aishwarya MS (Backend/Test)
4. Shahla Nigar (Backend/Test)

Agenda

1. Introduction to RevWorkForce
2. Objective
3. Tech Stack
4. Methodology
5. System Architecture
6. Design Patterns
7. Database Design
8. Core Features and Modules Employee/Manager/Admin
9. Security Features
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15. GitHub Walkthrough
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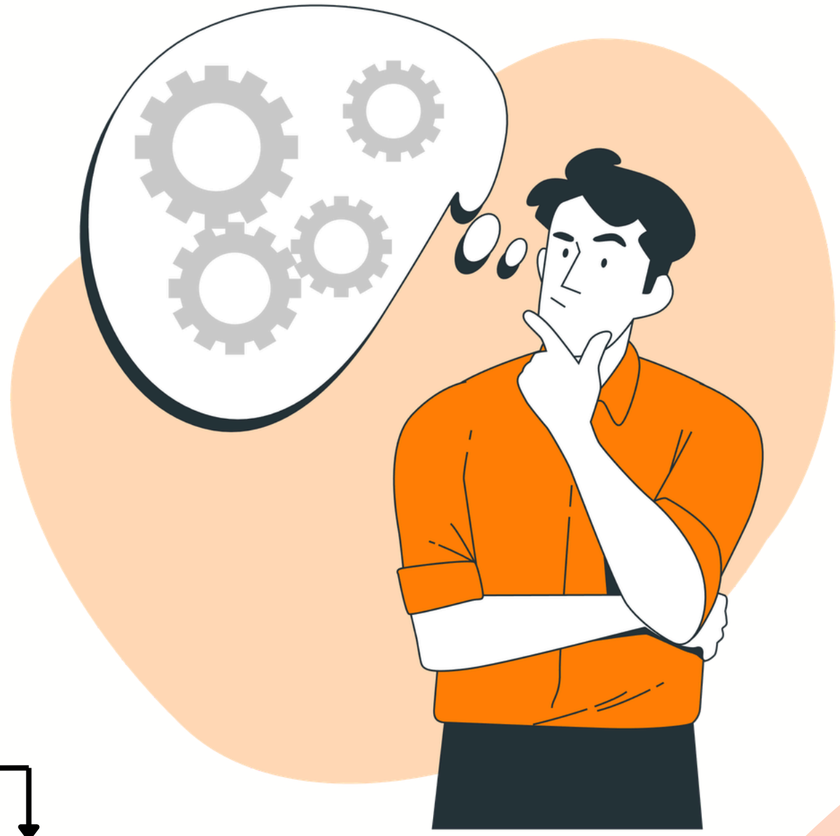
Introduction

What is RevWorkForce?

RevWorkforce is a full-stack monolithic Human Resource Management (HRM) web application designed to streamline employee management, leave tracking, and performance review processes.

The system implements role-based access control with three distinct user roles: **Employees**, **Managers**, and **Admins.**



Employee

(day-to-day HR tasks)

leave applications, profile updates, performance reviews

Manager

(Team Management)

team management, leave approvals, performance reviews

Admins

(system administration)

employee management, configuration

Objectives

Primary goal:

1. Develop a scalable, role-based HRM system
2. Implement secure authentication & authorization
3. Automate leave management workflows
4. Enable structured performance management
5. Centralize employee information
6. Implement enterprise-grade security
7. Provide real-time notifications and audit trails

Role-based security

14,000+ LOC

14 Tables

Tech Stack

Frontend

Thymeleaf • HTML • CSS • Bootstrap5
JavaScript • UI/Ux

Backend

Java 17+ • Spring Boot 3.2+
Spring Security • Spring Data JPA/Hibernate

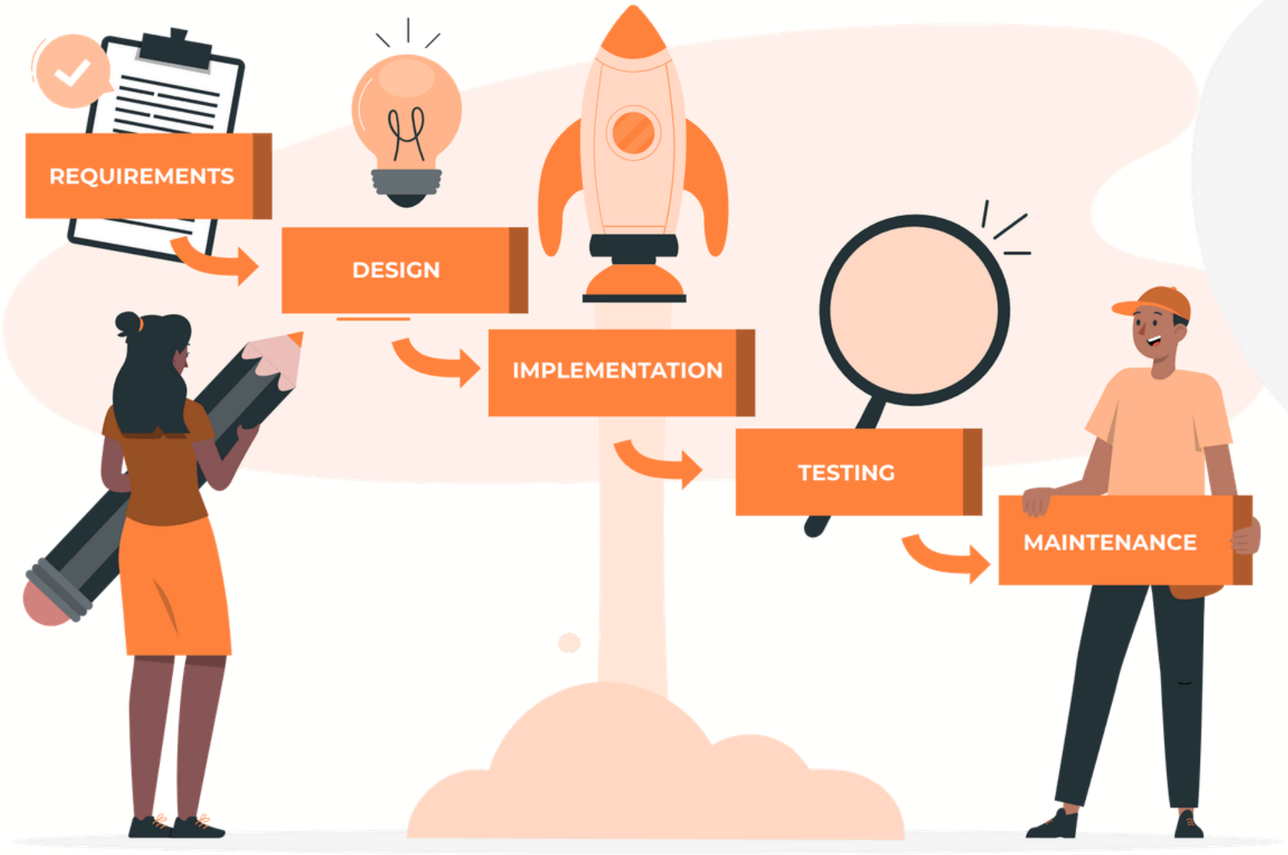
Database

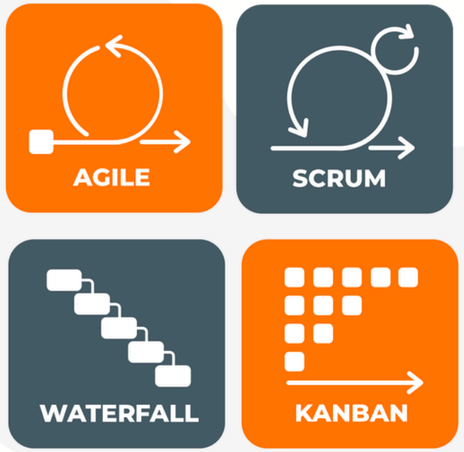
Oracle Database 21c XE
14 Tables • ACID Compliance

Tools

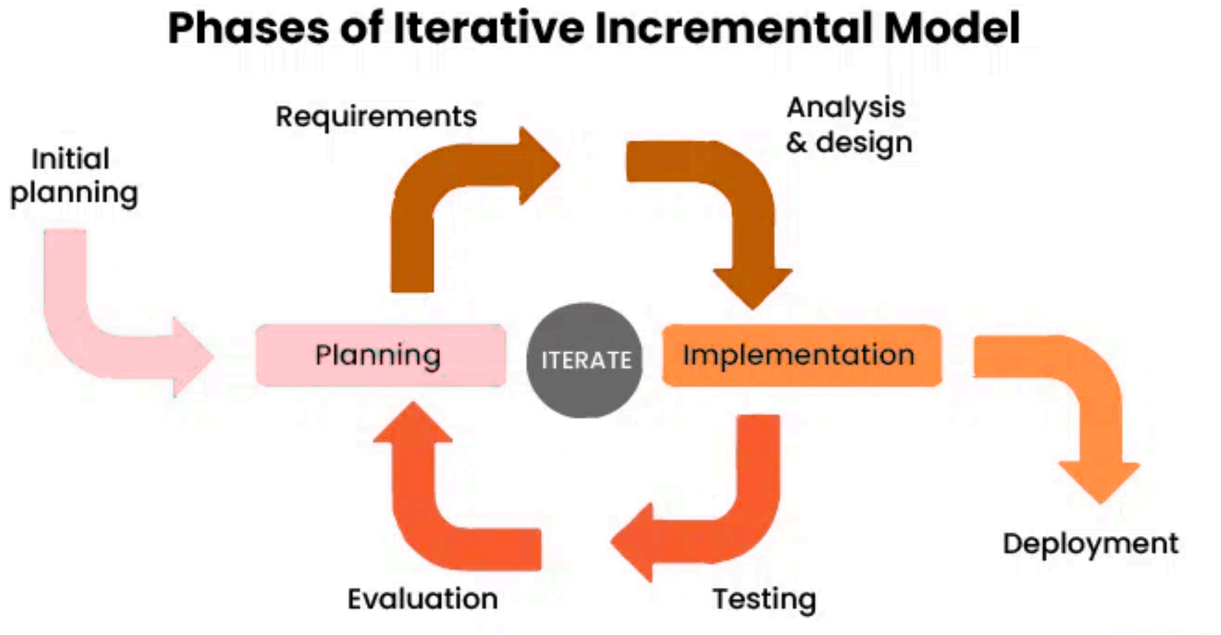
Maven • Jenkins • SonarQube
• JaCoCo • Git • GitHub • JUnit

Methodology





Methodology



Agile Practices

- Product Backlog
- Sprint Planning
- Burn Up Charts
- Kanban Board

- **Modular Nature:** The project has clearly defined modules (Authentication, Leave Management, Performance Management) that can be developed incrementally
- **Time Constraints**: 16-day timeline requires working software at each iteration
- **Risk Mitigation:** Early iterations help identify technical challenges early
- **Feedback Integration:** Each iteration can be tested and refined
- **Priority-Based Development:** Core features can be developed first, additional features later

Phases

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[PHASE 1] RevWorkForce - Foundation Setup & Core Architecture

#7 · MastanSayyad opened 2 weeks ago 1

This phase establishes the technical foundation of the RevWorkForce project.

complete functional development of all business modules and their UI implementation for the RevWorkforce system.

[Phase 2] - Module Development & UI Implementation

#76 · MastanSayyad opened 2 weeks ago

[PHASE 3] - Testing, CI/CD, Documentation & Finalization

#191 · MastanSayyad opened last week

focuses on final system validation, quality assurance, CI/CD implementation, documentation

System Architecture

Frontend Layer	Thymeleaf, Bootstrap 5, JavaScript
Controller Layer	Spring MVC, REST Controllers
Security Layer	Spring Security 6, RBAC, BCrypt Encryption, Session Mgmt
Service Layer	Business Logic, Transaction Management, Validation Rules
Repository Layer	Spring Data JPA, Hibernate ORM, Custom Query Methods
Database Layer	Oracle 21c XE, 14 Tables, Stored Procedures & Triggers

Design Patterns

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Singleton Pattern

DatabaseUtil, SessionManager
Ensures single instance

Service Layer

LeaveService, PerformanceService
Encapsulates business logic

Dependency Injection

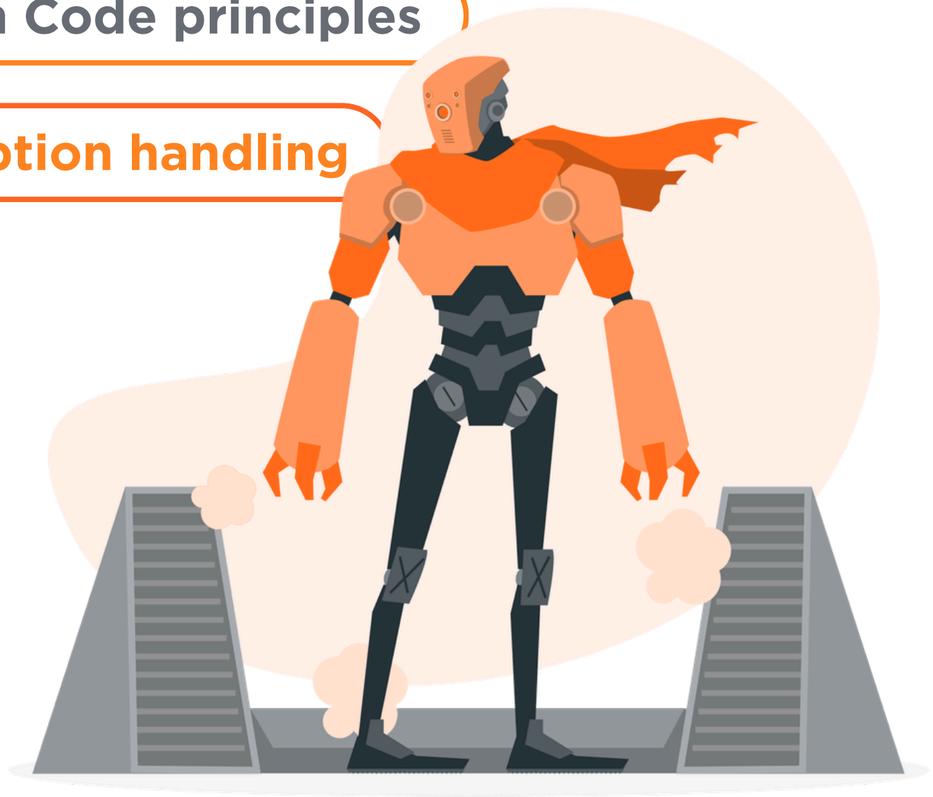
Services inject DAOs
Promotes loose coupling

DRY (Don't Repeat Yourself)

Meaningful naming conventions

Clean Code principles

Exception handling



Database Design

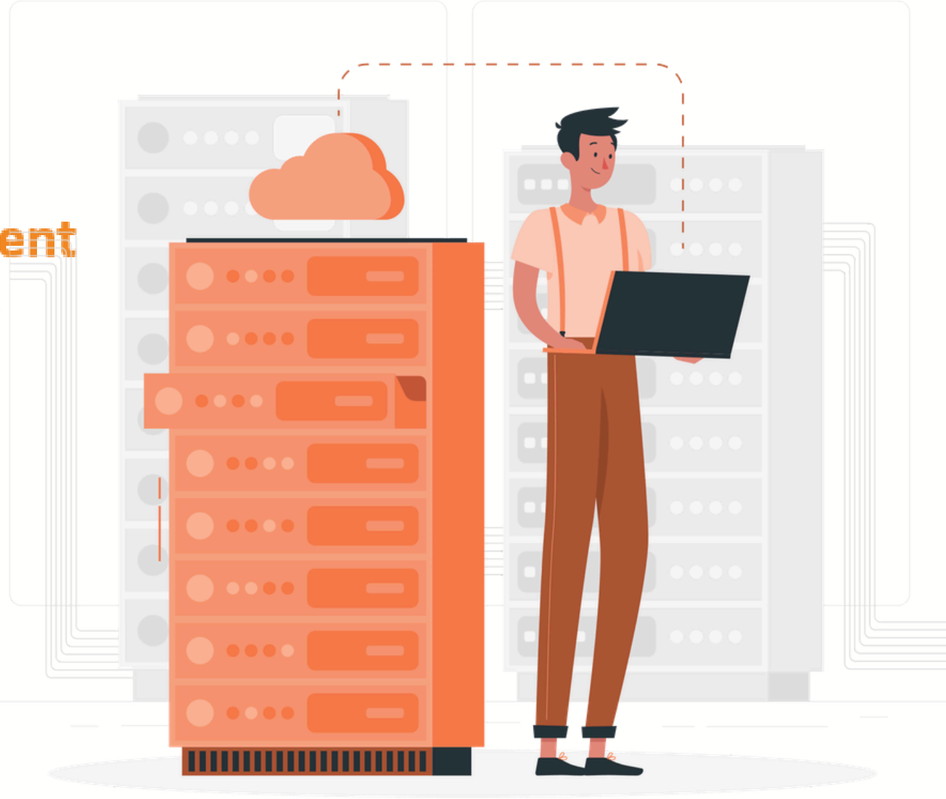
14 Tables | 6 Sequences | Normalized to 3NF

Employee Management
- EMPLOYEES
- DEPARTMENTS
- DESIGNATIONS
- ROLES

Performance Management
- PERFORMANCE_REVIEWS
- GOALS

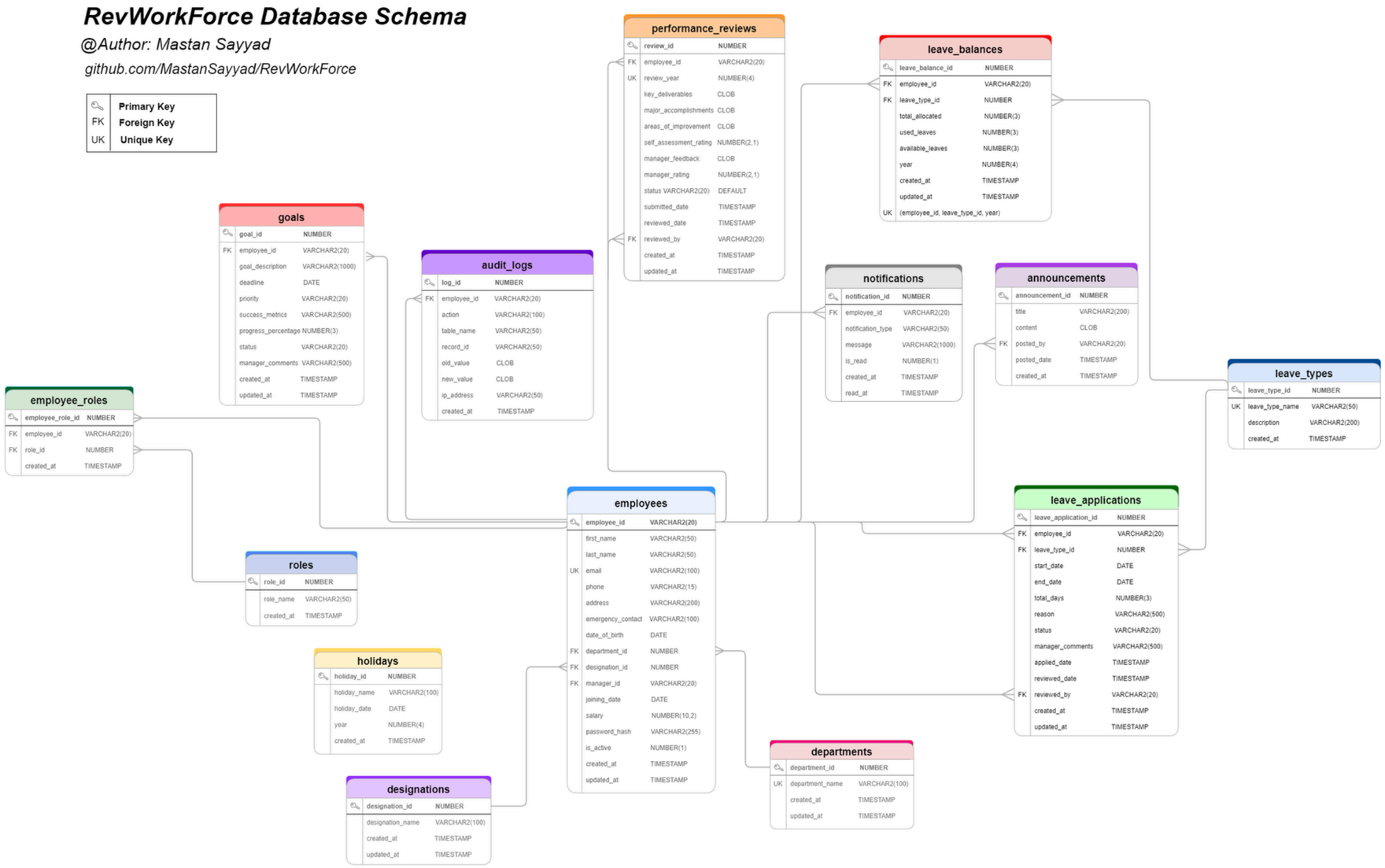
Leave Management
- LEAVE_APPLICATIONS
- LEAVE_TYPES
- LEAVE_BALANCES
- HOLIDAYS

System & Audit
- NOTIFICATIONS
- ANNOUNCEMENTS
- AUDIT_LOGS



Primary Keys (Sequences) | Foreign Keys | Indexes | ACID Compliance

Database Design



Core Features and Modules

Employee Management

- CRUD operations
- RBAC (3 roles)
- Department mapping
- Lifecycle tracking

Leave Management

- Multiple leave types
- Balance tracking
- Working days calc
- Approval workflow

Performance Mgmt

- Annual reviews
- Self-assessment
- Goal tracking
- Rating (1.0-5.0)

Notifications

- Real-time alerts
- Announcements
- Audit trail
- Birthday reminders

70% Time Reduction | 100% Accuracy | Real-time Visibility

Employee Features

Personal Dashboard

- Leave balance & upcoming events
- Quick actions for common tasks
- Goal progress visualization
- Recent notifications feed

Leave Management

- Apply leave with date picker
- View history & current status
- Real-time balance display
- Cancel pending requests

Performance & Goals

- Create and track personal goals
- Set deadlines and priorities
- Submit self-assessments
- View manager feedback & ratings

Announcements

- View company-wide posts
- Filter by priority level
- NEW badge for recent items
- Search functionality

Manager Features

Team Management

- View team member details
- Track attendance & availability
- Department roster overview
- Employee profile access

Leave Approvals

- Review pending requests
- Approve/reject with comments
- Team calendar view
- Conflict detection

Performance Reviews

- Review self-assessments
- Assign manager ratings 1-5
- Provide detailed feedback
- Track completion status

Goals Oversight

- Monitor team goals
- Provide guidance & comments
- Track completion rates
- Identify at-risk goals

Admin Features

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Employee Mgmt

- Full CRUD operations
- Assign roles & departments
- Set manager hierarchy
- Bulk operations support

System Config

- Manage holidays & leave types
- Configure departments
- Manage designations
- Company announcements

Audit Logs

- Complete activity tracking
- Filter by date/module/user
- Export CSV/Excel reports
- Immutable audit records

Analytics

- Leave utilization reports
- Performance statistics
- Goal completion metrics
- Department-wise analytics

Leave Management Workflow

1. Employee Applies	Select dates, type, reason
2. System Validates	Check balance, overlap, dates
3. Manager Notified	Real-time alert sent
4. Manager Reviews	Approve/reject with comments
5. Balance Updated	Auto-deduction on approval
6. Employee Notified	Status update sent

Security Features

Authentication

- Form-based login (Spring Security)
- BCrypt password encryption
- Remember Me functionality
- Account lockout after failures

Authorization (RBAC)

- Role-Based Access Control
- Method-level @PreAuthorize
- URL-based authorization rules
- 3 roles: Employee / Manager / Admin

Role Permissions

- Employee: personal data only
- Manager: team + approvals
- Admin: system-wide access
- Hierarchical permissions

Audit Logging

- Who, What, When, Where tracking
- Before/After update values
- 90-day retention policy

Ui/Ux Highlights

Glassmorphism Theme

- Semi-transparent card backgrounds
- Backdrop blur effects
- Gradient color schemes (Teal Trust)
- Consistent design language

Responsive Design

- Mobile-first Bootstrap 5 approach
- Works on phones, tablets, desktops
- Touch-friendly interface
- Adaptive navigation

Custom Error Pages

- 404 — Purple, floating
- 403 — Red, shaking
- 500 — Orange, spinning
- 400 — Blue, bouncing

UX Features

- Real-time form validation
- Toast notifications
- Confirm dialogs for destructive actions
- Search, filter & pagination

Live Demonstration

Admin Flow
3 min

1. Login as ADM001
2. Navigate to Employee Management
3. Create new employee
4. Assign roles & department
5. View audit logs

Employee Flow
3 min

1. Login as EMP001
2. View personal dashboard
3. Apply for leave
4. Create new goal
5. View announcements

Manager Flow
3 min

1. Login as MGR001
2. View pending leave requests
3. Approve leave with comments
4. Review team goals
5. Provide performance feedback

Total Demo Time: ~10 minutes • Real-time notifications • responsiveness • UI

Testing & Quality Assurance

76%+

Test Coverage
(JaCoCo)

GUARANTEED ✓

Unit Testing

- JUnit 5 + Mockito framework
- Service layer business logic
- Repository custom queries
- DTO conversions & utilities

Integration Testing

- End-to-end workflow tests
- H2 in-memory database tests

Coverage Formula: Service Tests → ~40% + Controller Tests → +20% = 60%+ Total ✓

Testing & Quality Assurance

RevWorkForce Public

Passed

Last analysis: 1 hour ago • **13k** Lines of Code • Java, XML

E 1	C 64	A 459	E 0.0%	69.7%	5.1%
Security	Reliability	Maintainability	Hotspots Reviewed	Coverage	Duplications

SonarQube Quality Gate: PASSED

0	0	< 50	< 5d	< 3%
Bugs	**Vulnerabilities**	**Code Smells**	**Tech Debt**	**Duplications**
Critical Issues	Security Issues	Maintainability	Days	Lines

A	A	A
Maintainability Rating	**Security Rating**	**Reliability Rating**

CI/CD Pipeline

GitHub Walkthrough

Repository Highlights

- Well-organized structure
- Comprehensive README
- DB setup scripts
- Maven configuration
- JaCoCo reports
- documentation
- Meaningful commits
- Issue Management

Development Workflow

- Feature branches
- Pull requests
- Main for releases
- GitHub Actions CI

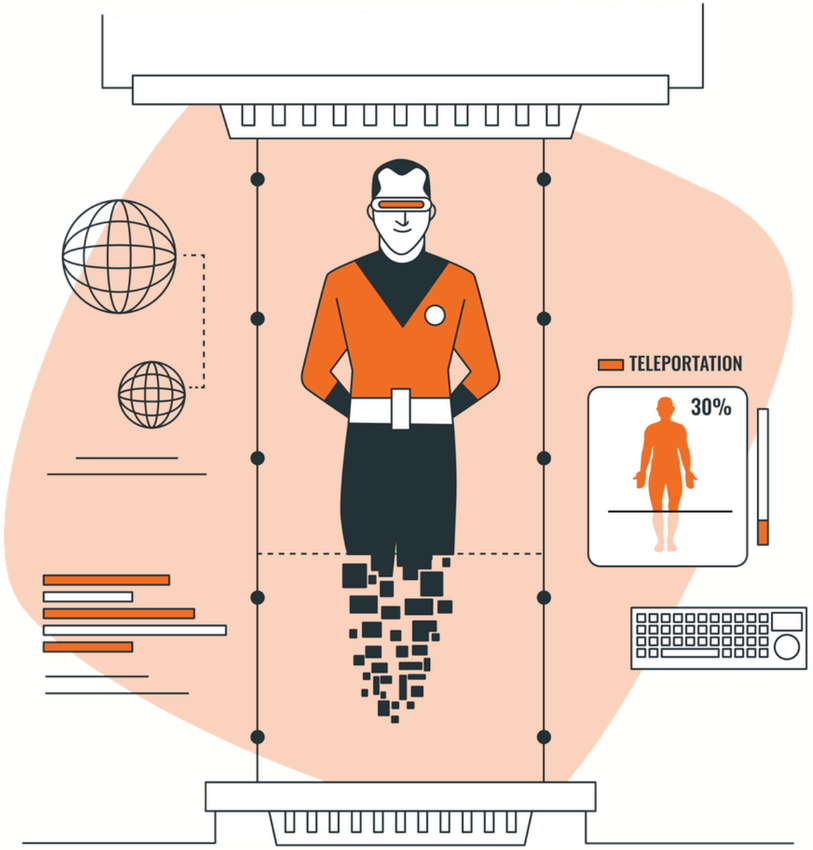
https://github.com/RevWorkForce-PES/P2-RevWorkForce

CHALLENGES FACED

01	Template Path Confusion	Problem: Mixed template locations causing 404 errors Fix: Standardized all paths to frontend/pages/	Consistent project structure is critical
02	Enum Mismatch	Problem: DB had NORMAL but Priority enum didn't Fix: Added NORMAL to Priority enum	Database and code must stay in sync
03	Duplicate Endpoint Conflicts	Problem: Multiple /manager/performance mappings Fix: Consolidated endpoints, fixed routing	Proper endpoint planning prevents conflicts
04	Error Page Rendering	Problem: Spring Boot 3 RFC 7807 JSON responses Fix: Disabled ProblemDetail, custom error controller	Framework defaults need customization
05	Username Display Bug	Problem: 'Loading...' shown instead of actual username Fix: Added fullName to all model attributes	Template variables must be populated

Merge Conflicts and Many More

Team Contribution



Conclusion

WHAT WE BUILT:

✓ Comprehensive HRM System
✓ Application quality code
✓ 14-table database schema
✓ 582+ test cases (60% coverage)
✓ Complete documentation

WHAT WE ACHIEVED:

✓ Solved real HR management challenges
✓ Implemented enterprise architecture
✓ Applied industry best practices
✓ Delivered production-ready system
✓ Created scalable foundation

Learnings: Software Engineering • Database Design • Security • Testing • Agile

Team Management • Version Control • Project Management • TDD • Code review• Documentation

Thank You

1. Mastan Sayyad (Team Lead)
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(Scan for Github Repo)

RevWorkForce Enterprise Human Resource Management System